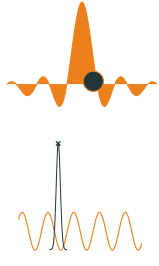
Non-Markovian collapse models can be recast as Bohmian-like theories



A. Tilloy and H. Wiseman

August 9th, 2022

Remembering Detlef Dürr

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Outline

1. Basics of collapse models
2. The flexibility of non-Markovian collapse models
3. The complexity of non-Markovian collapse models
4. Non-Markovian collapse models as Bohmian theories

Introduction to collapse models

Main idea of collapse models

Other names: [objective / spontaneous / dynamical] [collapse / reduction] [models / program]

Schrödinger equation + small non-linear perturbation

$$\frac{\mathrm{d}}{\mathrm{d}t}\psi_t = -\frac{i}{\hbar}H\psi_t + \varepsilon(\psi)\,,$$

H is the standard model Hamiltonian (or a non-relativistic simplification)

First consistent equation of this type around 1984

The easier Ghirardi-Rimini-Weber model

The GRW modification (1986)

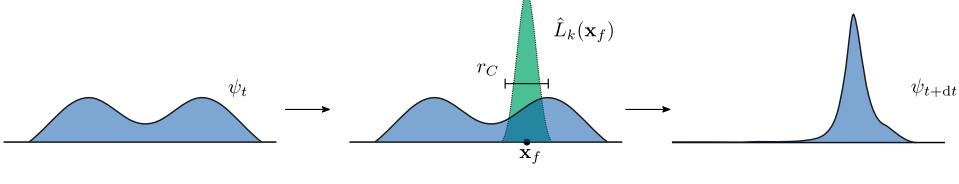
Every dt, with probability λdt particle k collapses around point x_f

$$\psi_t \longrightarrow \frac{\hat{L}_k(x_f)\psi_t}{\|\hat{L}_k(x_f)\psi_t\|} \text{ with proba } P(x_f) = \|\hat{L}_k(x_f)\psi_t\|^2$$

with an envelope $\hat{L}_k(x_f) = \frac{1}{(\pi r_C^2)^{3/4}} e^{-(\hat{x}_k - x_f)^2/(2r_C^2)}$.



GianCarlo Ghirardi
1935 - 2018



Why it works (solves the measurement problem)

Fixing *e.g.* $\lambda = 10^{-16}\mathrm{s}^{-1}$ (historical value) :

1. An electron collapses every 300 million years
2. A cat made of $\simeq 10^{28}$ electrons is localized up to r_c in less than a picosecond

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In brief: one can semi-rigorously derive the measurement postulate by studying the stochastic dynamics of measurement devices

Microscopic degrees of freedom (spin, photon, etc.) do not collapse because of their intrinsic dynamics, but when they are coupled to something macroscopic

The master equation

Empirical content of GRW

One can only measure frequencies $\pi_k = \langle\psi|\hat{\Pi}_k|\psi\rangle$, averaged over jumps not knowable *a priori* $\bar{\pi}_k = \mathbb{E}\left[\pi_k\right]$

$$\bar{\pi}_k = \mathbb{E}\left[\langle\psi|\hat{\Pi}_k|\psi\rangle\right] = \mathrm{tr}\left(\hat{\Pi}_k\,\mathbb{E}\left[|\psi\rangle\langle\psi|\right]\right) = \mathrm{tr}\left(\rho\,\hat{\Pi}_k\right).$$

$\Longrightarrow$ all experimentally falsifiable predictions of the model are in $\rho = \mathbb{E}\left[|\psi\rangle\langle\psi|\right]$

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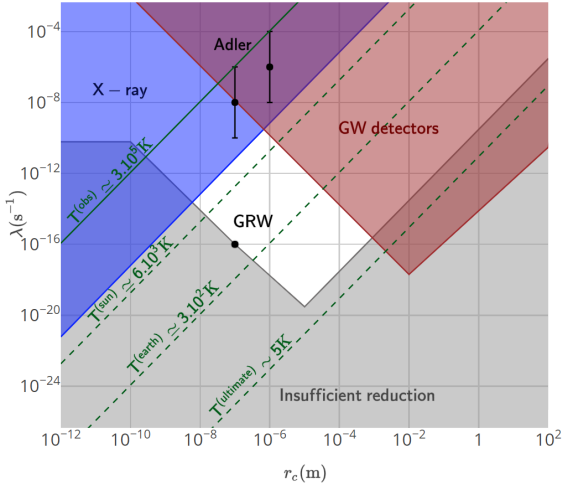
Master equation of GRW

Collapse probabilities are chosen exactly so that $\mathbb{E}$ removes the non-linearity

$$\frac{\mathrm{d}}{\mathrm{d}t}\hat{\rho}_t = -\frac{i}{\hbar}[\hat{H}, \rho_t] + \lambda \sum_{k=1}^{N} \left\{ \int \mathrm{d}x_f \hat{L}_k(x_f)\rho_t \hat{L}_k(x_f) \right\} - \hat{\rho}_t$$

Experimental consequences

1. Loss of interferences for big molecules
2. Matter slowly heats up
3. Stuff vibrates
4. Photons spontaneously get emitted



Some candidates

1) Markus Arndt's experiments 2) Neptune / neutron stars 3) Mirors of LISA pathfinder 4) Germanium crystals in Gran Sasso

3 levels of description

Ontology

"What the theory says about the world"

e.g. flashes (x_f, t_f)

The wave-function

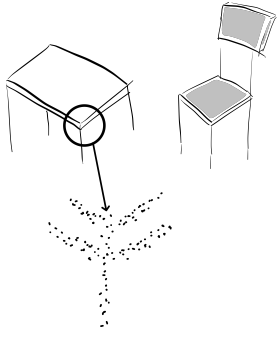
"An intermediary object"

$$\frac{\mathrm{d}}{\mathrm{d}t}\psi_t = -\frac{i}{\hbar}H\psi_t + \varepsilon(\psi)$$

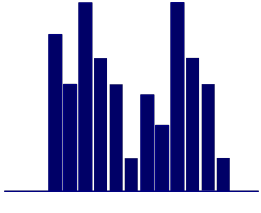
The empirical content

"Quantitative testable predictions of the theory"

$$\partial\rho_t = \mathscr{L}(\rho_t)$$



ψ?



Playing with representations

Averaging and unraveling

For $\rho = \mathbb{E}[|\psi\rangle\langle\psi|$ where $|\psi\rangle$ evolves stochastically

- **Averaging:** $|\psi\rangle \longrightarrow \rho$
- **Unraveling:** $\rho \longrightarrow |\psi\rangle$ many, some not inducing collapse

Playing with representations

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Tracing out and dilating

For $\rho = \mathrm{tr}[|\Psi\rangle\langle\Psi|]$ where $|\Psi\rangle$ evolves unitarily in a larger Hilbert space

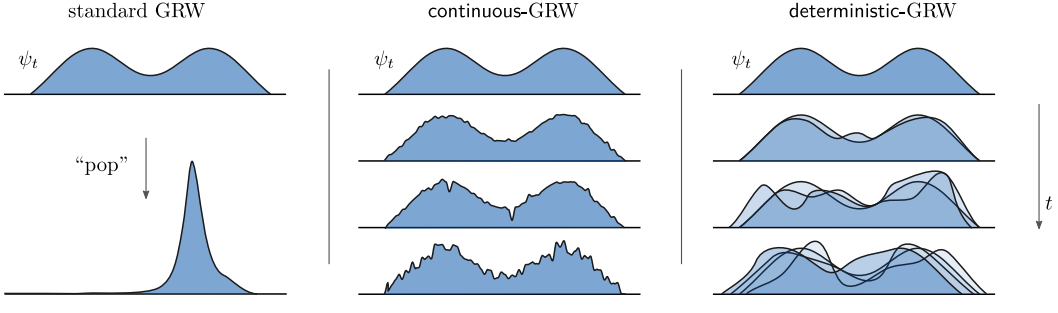
- **Tracing out:** $|\Psi\rangle \longrightarrow \rho$
- **Dilating:** $\rho \longrightarrow |\Psi\rangle$

For a large class of master equations, including all non-relativistic collapse models, explicit unravelings and dilations are known

Testing Quantum theory?

For example for **GRW**. On can construct empirically equivalent models that are:

- stochastic but continuous, and that do not collapse cats
- deterministic but with an added peculiar dark sector in the Standard Model



see *The subtle sound of quantum jumps* arXiv:2007.15420

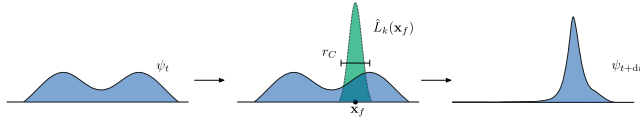
for GRW, deterministic-GRW is not an appealing description: dark sector memoriless

The flexibility of Non-Markovian collapse models

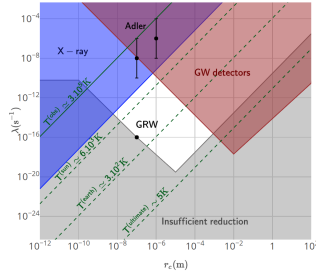
The tempting non-Markovianity

Two birds one stone:

- Allows covariant space-time regulation $\rightarrow$ relativity



- Tames the brutal empirical side effects of collapse



Introduced by [Strunz, Gisin, Diósi] and [Bassi, Ghirardi, Adler]

Recipe for collapse model construction

Build a collapse model with clear empirical content in a **3 simple steps**



1. Choose a nice unitary model of Fermions and Bosons, minimally coupled
2. Trace out the Bosons to get a (wildly non-Markovian) open system evolution for Fermions
3. Unravel this open evolution into a collapse evolution for Fermions

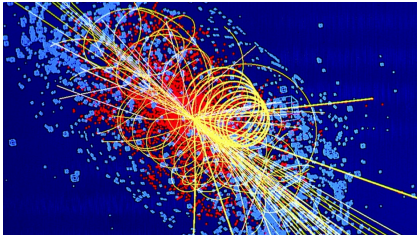
Benefits

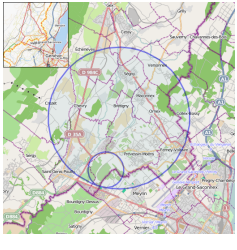
- Transparent predictions
- One can choose an unraveling yielding collapse

Hiding a collapse model inside the standard model

The **standard model** is just an instantiation of quantum mechanics

1. Hilbert space $\mathscr{H} = \mathscr{H}_f \otimes \mathscr{H}_b$ (the bosons and fermions)
2. Hamiltonian $H = H_f + H_b + H_{\text{int}}$ (the dynamics)

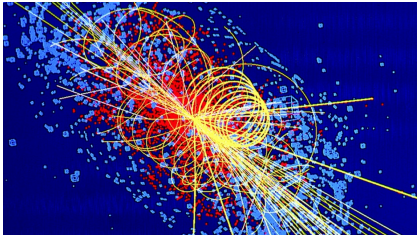


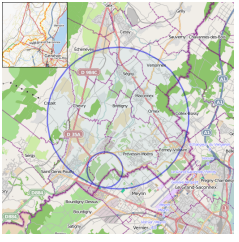


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Apply previous recipe:

- Start from $|\Psi_{fb}\rangle$
- Trace out $\mathscr{H}_b$ to get ρ_f
- Unravel ρ_f into $|\psi_f\rangle$

done on the example of Yukawa theory in `arXiv:1702.06325`

The complexity of non-Markovian collapse models

Collapse models with colored noise: linear evolution

Linear differential equation

$$\frac{\mathrm{d}}{\mathrm{d}t}|\phi_w(t)\rangle = \Bigg[-iH + \underbrace{\sqrt{\gamma}\, w_i(t) A_i}_{\text{noise drive}} - \underbrace{2\sqrt{\gamma}\, A_i \int_0^t \mathrm{d}s\, D_{ij}(t,s) \frac{\delta}{\delta w_j(s)}}_{\text{weird necessary memory term}} \Bigg] |\phi_w(t)\rangle,$$

with **colored** noise field: $\mathbb{E}\left[w_i(t) w_j(s)\right] = D_{ij}(t,s)$

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- **Master equation**: For $\rho_t = \mathbb{E}\Big[\, |\phi_w(t)\rangle\langle\phi_w(t)| \,\Big]$, on has $\rho_t = \Phi_t \cdot \rho_0$ with Φ_t Completely Positive Trace Preserving (CPTP)
- **Dilation**: the master equation admits a well known unitary dilation with a **non-Markovian** bosonic bath

Collapse models with colored noise: non-linear

Normalization

$$|\psi_w(t)\rangle = \frac{1}{\sqrt{\langle\phi_w(t)|\phi_w(t)\rangle}}|\phi_w(t)\rangle$$

but now $|\psi_w(t)\rangle$ does not yield a linear master equation.

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Noise tilt

Restore the linear master equation by biasing the noise field

$$w_j^{[t]}(s) = w_j(s) + 2\sqrt{\gamma}\int_0^t \mathrm{d}\tau\, D_{ij}(\tau, s)\langle A_i\rangle_\tau,$$

Full collapse model = linear equation + normalization + noise drift

Salient points to remember

Generic collapse models *i.e.* with colored noise, are more subtle than they seem. In particular, 2 difficulties:

1. **Memory term** :

$$2\sqrt{\gamma}\, A_i \int_0^t \mathrm{d}s\, D_{ij}(t,s) \frac{\delta}{\delta w_j(s)} |\phi_w(t)\rangle$$

   necessary to maintain consistency but extremely not intuitive

2. **Cooking / noise redefinition** :

$$\forall t \geqslant 0, \quad w_j^{[t]}(s) = w_j(s) + 2\sqrt{\gamma} \int_0^t \mathrm{d}\tau\, D_{ij}(\tau, s) \langle A_i \rangle_\tau,$$

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$\rightarrow$ Equivalent in dBB : **conditioning** for 1 and **guiding equation** for 2

Non-Markovian collapse models as Bohmian theories

Quantum 5, 594 (2021) – arXiv:2105.06115

Weak inclusion

Empirically: collapse $=$ Bohm on system $+$ bath

For a given collapse model on a system S, there exists a bath B such that Bohmian mechanics on $\{S + B\}$ has the same empirical content.

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[Gisin 1989, 1991]

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   `[Gisin 1989, 1991]`
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3. The master equation can be dilated into a unitary evolution

$$\exists\ H \text{ and } |\Psi_0\rangle = |\psi_0\rangle\otimes|\text{aux}\rangle \text{ s.t. } \partial_t|\Psi_t\rangle = -iH|\Psi_t\rangle \text{ and } \mathrm{tr}_B\Big[\, |\Psi_t\rangle\langle\Psi_t| \,\Big] = \rho_t$$

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4. This dilated unitary dynamics can be Bohmianized

Strong inclusion

Metaphysically: collapse = Bohm on system + bath

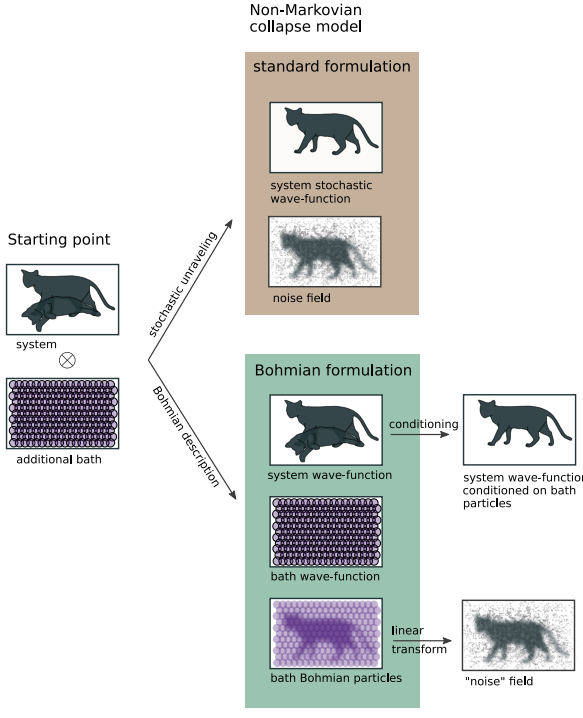
For a given collapse model on a system S, there exists a bath B such that Bohmian mechanics on $\{S + B\}$ **contains the exact same objects**

- The Bohmian wave-function on $S + B$ conditioned on bath hidden variables **is** the collapse model stochastic wave function:

$$|\psi^{\text{collapse}}\rangle = |\psi_C^{\text{Bohm}}(x_{\text{bath}})\rangle$$

- The Bohmian hidden variables of the bath are in one to one correspondence with the collapse noise

$$x_{\text{bath}}^{\text{Bohm}} \underset{\text{linear transform}}{\longleftrightarrow} \textbf{collapse noise } w$$


Non-Markovian
collapse model
standard formulation
system stochastic
wave-function
noise field
Starting point
system
additional bath
stochastic unraveling
Bohmian description
Bohmian formulation
system wave-function
conditioning
system wave-function
conditioned on bath
particles
bath wave-function
linear
transform
bath Bohmian particles
"noise" field

Precise formulation

Collapse model

For states $|\psi\rangle$ in a Hilbert space $\mathscr{H}_s$, we are given a collapse model "collapsing" a set of operators A_i, $i = 1 \cdots d$, with a colored noise field $w_i(t)$:

$$\partial_t |\psi_w\rangle = \underbrace{-iH|\psi\rangle}_{\text{linear QM}} + \underbrace{f(A_i, w_i, \psi_w)\,|\psi_w\rangle}_{\text{collapse term}}$$

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Bohmian theory on extended space

One can find a bath Hilbert space $\mathscr{H}_b$ made of interacting harmonic oscillators $\{x_{i,\omega}\}$ indexed by $i = 1 \cdots d$ and a frequency $\omega \in \mathbb{R}$, and a (reasonable) interaction Hamitonian H_{int} such that:

1. $|\Psi(0)\rangle = |\psi(0)\rangle \otimes |0\rangle \in \mathscr{H}_{\text{total}} = \mathscr{H}_s \otimes \mathscr{H}_b$
2. $|\psi^C(t)\rangle = \langle\{x_{i,\omega}(t)\}|\Psi_t\rangle \ \in \mathscr{H}_s$
3. For $x_{i,\omega}(0)$ drawn with the **Born rule**, $|\psi^C\rangle$ has the same law as $|\psi_w\rangle$

Comments

1. Technically, the proof uses an "old" result
   $\rightarrow$ [Gambetta & Wiseman 2003]
2. The result is valid for continuous collapse models with **non-white noise**
3. The result is still valid for continuous collapse models with white noise, but unimpressive – Bohmian description far more complicated
4. The result is obtained in a bruteforce way, with an explicit non-trivial computation
5. Applying to QED $\rightarrow$ Struyve Westman proposal of Bohmian field theory for bosons

Intuition for the proof of the equivalence

Linear collapse equation = conditioning on **fixed** Bohmian hidden variables

$$\frac{\mathrm{d}}{\mathrm{d}t}|\phi_w(t)\rangle \underset{\text{collapse}}{:=} \left[-iH + \sqrt{\gamma}\, w_i(t) A_i - 2\sqrt{\gamma}\, A_i \int_0^t \mathrm{d}s\, D_{ij}(t,s) \frac{\delta}{\delta w_j(s)} \right] |\phi_w(t)\rangle$$

$$|\phi_w(t)\rangle \underset{\text{Bohm}}{:=} \langle w|\Psi\rangle_{S+B} \quad \underset{\text{conditioning}}{\longleftarrow} \quad |\Psi\rangle_{S+B} \quad \text{complete wavefunction}$$

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Non-trivial continuous noise drift = Bohmian guiding law

$$w_j^{[t]}(s) \underset{\text{collapse}}{:=} w_j(s) + 2\sqrt{\gamma} \int_0^t \mathrm{d}\tau\, D_{ij}(\tau, s) \langle A_i \rangle_\tau$$

$$\frac{\mathrm{d}}{\mathrm{d}t} w_j^{[t]}(s) \underset{\text{Bohm}}{:=} \mathcal{V}(\psi, w)$$

Random lessons & questions

1. We should be careful of distinguishing **quantum theory** from its instantiation in the **Standard Model**
2. One approach to foundations can be mathematically reduced to the other – beyond just the empirical content
3. The distinction between a **deterministic** and a **stochastic** theory is inevitably blurry
   - In the nM collapse model description randomness is progressively revealed
   - In the Bohmian mechanics rewriting, it is in the initial condition
4. **Guiding laws** are ubuquitous: Is there another way to construct realist interpretations without them